Functions

ANSWER KEY

Evaluating functions

1 If $f(x) = 3x - 4$, find $f(5)$.
$f(5) = 3(5) - 4 = 11$.

2 If $f(x) = x^2 - 2x + 1$, find $f(-2)$.
$f(-2) = 4 + 4 + 1 = 9$.

3 If $g(x) = \sqrt{x+5}$, find $g(4)$.
$g(4) = \sqrt{9} = 3$.

4 If $h(x) = \frac{2x}{x-1}$, find $h(3)$.
$h(3) = \frac{6}{2} = 3$.

Domain and range

5 State the domain of $f(x) = \sqrt{x-3}$.
$x - 3 \geq 0$, so the domain is $x \geq 3$.

6 State the domain of $f(x) = \frac{1}{x-5}$.
All real numbers except $x = 5$ (where the denominator is zero).

7 State the domain of $f(x) = \sqrt{7-x}$.
$7 - x \geq 0$, so the domain is $x \leq 7$.

8 State the range of $f(x) = x^2 + 2$.
Since $x^2 \geq 0$ for all real x, the range is $f(x) \geq 2$.

Transformations: shifts

9 The graph of $y = x^2$ is shifted 3 units right and 2 units down. Write the new equation.
$y = (x-3)^2 - 2$.

10 The graph of $y = |x|$ is shifted 4 units left and 1 unit up. Write the new equation.
$y = |x+4| + 1$.

11 Describe the transformation that takes $y = x^2$ to $y = (x+2)^2 - 5$.
A shift 2 units left and 5 units down.

Transformations: reflections and stretches

12 The graph of $y = x^2$ is reflected over the x-axis and stretched vertically by a factor of 3. Write the new equation.
$y = -3x^2$.

13 Describe the transformation that takes $y = \sqrt{x}$ to $y = 2\sqrt{x}$.
A vertical stretch by a factor of 2.

14 Describe the transformation that takes $y = x^2$ to $y = -(x-1)^2 + 4$.
A reflection over the x-axis, then a shift 1 unit right and 4 units up.

Composite functions

15 Let $f(x) = 2x + 1$ and $g(x) = x^2 - 3$. Find $(f \circ g)(2)$.
$g(2) = 4 - 3 = 1$. $f(1) = 2(1) + 1 = 3$.

16 Using the same f and g, find $(g \circ f)(2)$.
$f(2) = 5$. $g(5) = 25 - 3 = 22$.

17 Using the same f and g, find a general formula for $(f \circ g)(x)$.
$f(g(x)) = 2(x^2 - 3) + 1 = 2x^2 - 6 + 1 = 2x^2 - 5$.

18 Using the same f and g, find a general formula for $(g \circ f)(x)$.
$g(f(x)) = (2x+1)^2 - 3 = 4x^2 + 4x + 1 - 3 = 4x^2 + 4x - 2$.

Inverse functions

19 Find the inverse of $f(x) = 3x - 2$.
$y = 3x - 2 \Rightarrow x = 3y - 2 \Rightarrow y = \frac{x+2}{3}$. So $f^{-1}(x) = \frac{x+2}{3}$.

20 Find the inverse of $f(x) = \frac{x-4}{5}$.
$y = \frac{x-4}{5} \Rightarrow 5y = x - 4 \Rightarrow x = 5y + 4$. So $f^{-1}(x) = 5x + 4$.

21 Show that $f(x) = 2x + 6$ and $g(x) = \frac{x-6}{2}$ are inverses by computing $f(g(x))$.
$f(g(x)) = 2\left(\frac{x-6}{2}\right) + 6 = (x-6) + 6 = x$, confirming they are inverses.

22 Find the inverse of $f(x) = x^3 + 1$.
$y = x^3 + 1 \Rightarrow x = y^3 + 1 \Rightarrow y = \sqrt[3]{x-1}$. So $f^{-1}(x) = \sqrt[3]{x-1}$.

Piecewise functions

23 A function is defined by $f(x) = x + 2$ for $x < 1$, and $f(x) = 3x - 1$ for $x \geq 1$. Find $f(-2)$.
Since $-2 < 1$, use the first rule: $f(-2) = -2 + 2 = 0$.

24 Using the same piecewise function, find $f(1)$.
Since $1 \geq 1$, use the second rule: $f(1) = 3(1) - 1 = 2$.

25 Using the same piecewise function, find $f(4)$.
Since $4 \geq 1$, use the second rule: $f(4) = 3(4) - 1 = 11$.

Even and odd functions

26 Determine whether $f(x) = x^2 - 3$ is even, odd, or neither.
$f(-x) = (-x)^2 - 3 = x^2 - 3 = f(x)$, so f is even.

27 Determine whether $f(x) = x^3 - x$ is even, odd, or neither.
$f(-x) = -x^3 + x = -(x^3 - x) = -f(x)$, so f is odd.

28 Determine whether $f(x) = x^2 + x$ is even, odd, or neither.
$f(-x) = x^2 - x$, which equals neither $f(x)$ nor $-f(x)$, so f is neither even nor odd.

Increasing, decreasing, and average rate of change

29 Find the average rate of change of $f(x) = x^2$ from $x = 1$ to $x = 4$.
$\frac{f(4) - f(1)}{4 - 1} = \frac{16 - 1}{3} = \frac{15}{3} = 5$.

30 Find the average rate of change of $f(x) = 2x^2 - 3x$ from $x = 0$ to $x = 3$.
$f(3) = 18 - 9 = 9$, $f(0) = 0$. $\frac{9 - 0}{3} = 3$.

31 State the interval on which $f(x) = (x - 2)^2$ is decreasing.
The vertex is at $x = 2$ and the parabola opens upward, so f is decreasing for $x < 2$.

Operations on functions

32 Let $f(x) = x + 3$ and $g(x) = x^2 - 1$. Find $(f + g)(x)$.
$(f + g)(x) = (x + 3) + (x^2 - 1) = x^2 + x + 2$.

33 Using the same f and g, find $(f \cdot g)(2)$.
$f(2) = 5$, $g(2) = 3$. $(f \cdot g)(2) = 5 \times 3 = 15$.

34 Using the same f and g, find $(f/g)(3)$.
$f(3) = 6$, $g(3) = 8$. $(f/g)(3) = \frac{6}{8} = \frac{3}{4}$.

Reading a function graph

35 The graph of $y = f(x) = x^2 - 4$ is shown.

a State the zeros of f. $x = -2$ and $x = 2$ (where the curve crosses the x-axis).

b State the domain and range of f. Domain: all real numbers. Range: $y \geq -4$.

Word problems: functions in context

36 This question has two parts. A company's daily profit (in dollars) from selling x units is modeled by $P(x) = -2x^2 + 80x - 150$.

a Evaluate $P(10)$ to find the profit from selling 10 units.
$P(10) = -2(100) + 800 - 150 = -200 + 800 - 150 = 450$ dollars.

b Find the number of units x that maximizes profit (using the vertex of the parabola), and state the maximum profit. The vertex is at $x = -\frac{80}{2(-2)} = 20$ units. $P(20) = -2(400) + 1600 - 150 = 650$ dollars — the maximum profit.

37 This question has two parts. The Fahrenheit conversion of a Celsius temperature is $F(C) = \frac{9}{5}C + 32$. Meanwhile, the temperature in a city t hours after 6 a.m. is modeled by $C(t) = -0.5t^2 + 6t + 10$ (in degrees Celsius), for $0 \leq t \leq 12$.

a Find $C(4)$, the Celsius temperature 4 hours after 6 a.m.
$C(4) = -0.5(16) + 24 + 10 = -8 + 24 + 10 = 26$°C.

b Find $(F \circ C)(4)$, the Fahrenheit temperature at that time.
$(F \circ C)(4) = \frac{9}{5}(26) + 32 = 46.8 + 32 = 78.8$°F.